ITCS216 Project Phase I

Subtitle: Jett Fitness

Section: 1
Group: Group 7
Date: Sep 27, 2026

Team Members (Sorted by Student ID)

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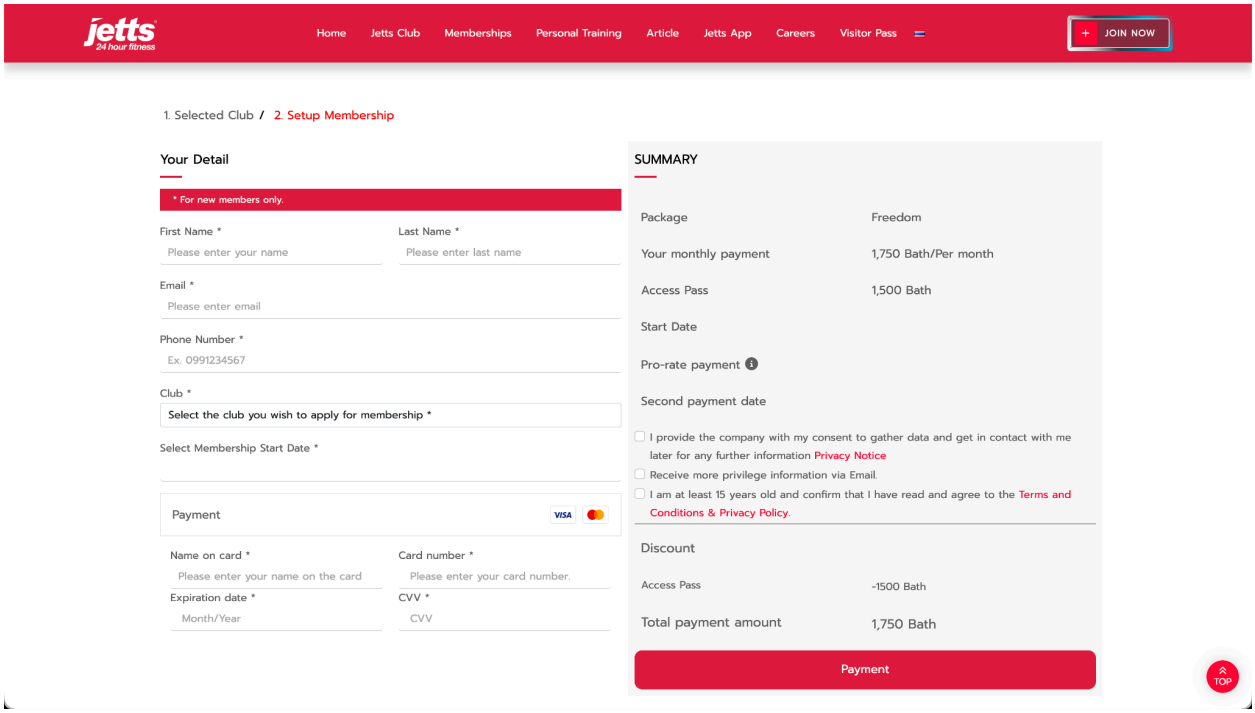
1. Introduction: Business Domain Overview

Jetts Fitness is an Australia-based chain of fitness centers founded in Queensland in 2007, operating across countries including New Zealand, the Netherlands, the UK, India, and Thailand. The gym provides 24/7 access, no lock-in contract, flexibility, experienced staff, and free monthly member classes. Facilities are categorized into Red (standard facilities) and Black (larger, luxury tier) gyms, accessible via electronic member cards; members can access the same-tier gym by using the electronic member card. Jetts offers tiered memberships including Freedom (1,750 Baht/month), Flexi (1,650 Baht/month, 6-month minimum), and Stay Fit (1,550 Baht/month, 12-month minimum). The project scope encompasses core administrative operations across member onboarding, monthly billing, and transaction processing.

2. Workflow and Operational Activities

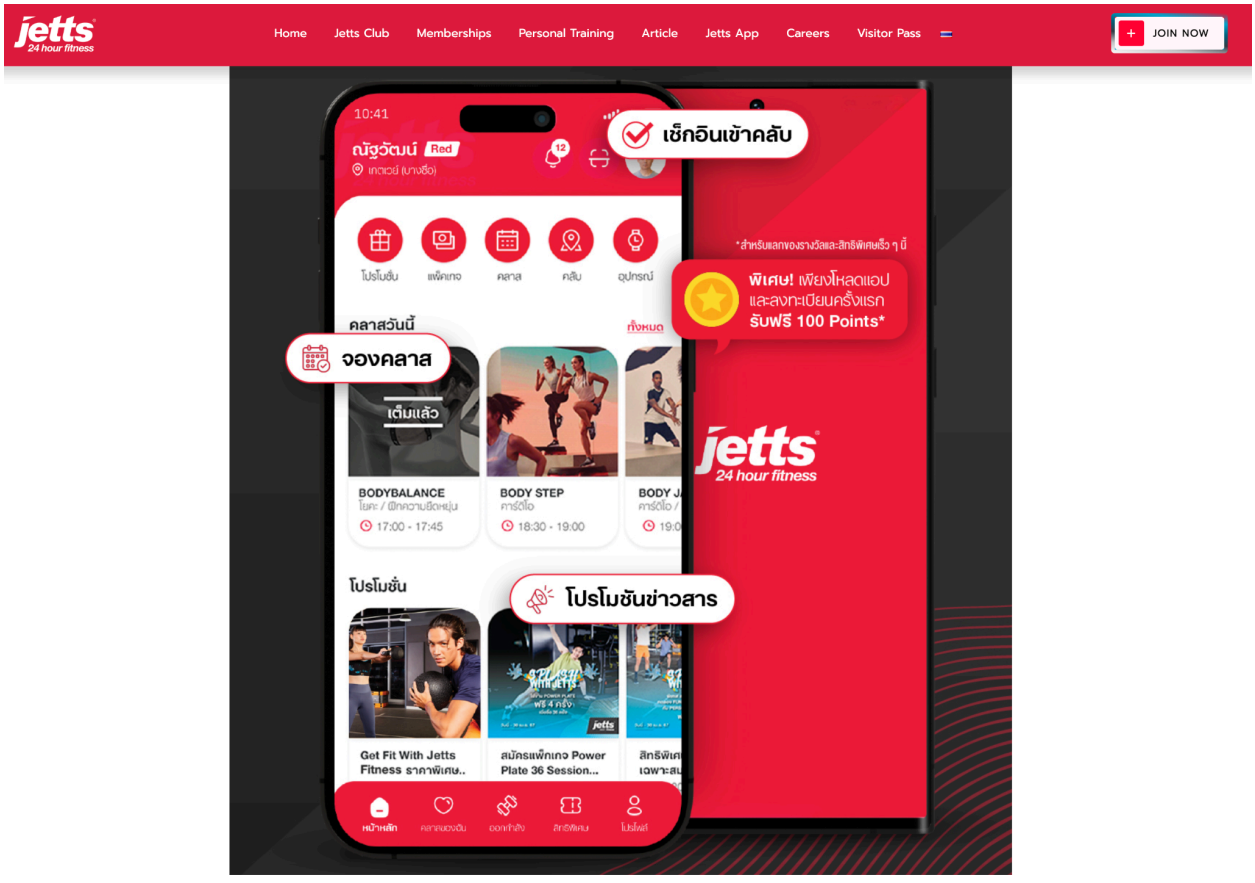
2.1 Membership Registration

1. Who initiates the action?
    a. The customer and the Front-Desk Staff are the action initiators.
2. What data is captured or updated?
    a. Customer's personal identity information, emergency contacts, payment method credentials, selected Membership plan, registration timestamp.
3. What state change occurs in the system?
    a. User/Member State: Customer -> Member
    b. Access Control State: No access -> Active Access.
4. Example[4]:
    a. A customer walks into a club branch, asking to become a member; the front-desk staff will give a membership registration form. The customer needs to fill in personal identity information, emergency contacts, payment method, selected membership plan, and signature. Then the staff will log that information into the system, and once the payment is confirmed, an electronic card will be provided to the member. The member can use it to access all the same-tier gym branches.



2.2 Class Registration (Addition)

1. Who initiates the action?
    a. The member (via Jetts APP for reservation) and Group Class Instructor / Front-Desk Staff (for attendance verification)
2. What data is captured or updated?
    a. Booking ID, Class ID, Instructor ID, scheduled session timestamp, member reservation status (e.g., Confirmed, Waitlisted, Attended, No-Show), and remaining seat count.
3. What state change occurs in the system?
    a. Available Slots -> Decremented Slots
    b. Booking Status: Null -> Confirmed -> Attended (or Cancelled/No-Show)
4. Example[8]:
    a. A member reserves a spot in a 6:00 PM Boxing class via the app. The system verifies the class has available seats, reserves seat #7, reduces the remaining seats by one, and assigns Booking ID #xxx. At 5:50 PM, the instructor checks in the member, moving the booking status from Confirmed to Attended.



2.3 Personal Training Package Purchase & Session Redemption(Addition)

1. Who initiates the action?
    a. The member (purchasing and scheduling) and the Personal Trainer (logging session completion)
2. What data is captured or updated?
    a. PT (Personal Trainer) Package ID, assigned Trainer ID, consumed sessions, remaining session balance, session completion timestamp, and member sign-off verification.
3. What state change occurs in the system?
    a. PT Credit Balance: Remaining Sessions -> Decrement Session. Once a session is booked and completed, the balance should decrease by the number of sessions
4. Example[7]:
    a. A member completes a 50-minute session with their assigned personal trainer. The trainer submits the session record through the front-desk staff. The system deducts 1 session from the member's 10-session package balance, and credits the trainer's commission log.

Group training package

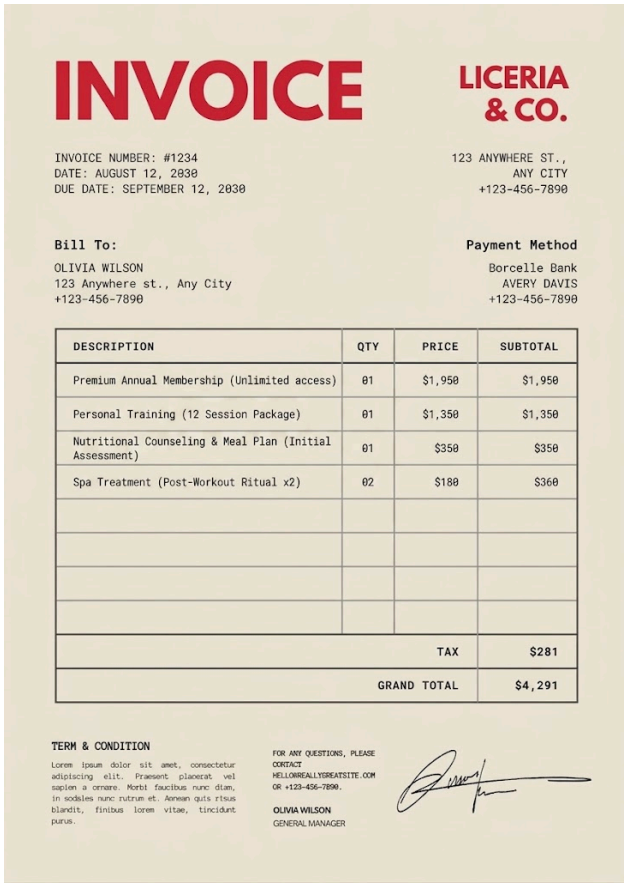
Jetts red package	Price	Rate	Expiration
10SS	16,000	1,600	60 Days

Boxing trainging package

Jetts red package	Price	Rate	Expiration
12SS	13,800	1,150	90 Days
24SS	25,000	1,042	150 Days
36SS	33,600	1042	180 Days
50SS	42,900	858	240 Days
50SS	42,900	858	240 Days

2.4 Monthly Invoice

1. Who initiates the action?
    a. Billing System, the automated system executing at the start of a billing cycle.
2. What data is captured or updated?
    a. Invoice number, issue date, billing cycle period, line items, total amount due, and payment due date.
    b. The member's account balance ledger is increased by the invoice amount, and the next billing timestamp is updated.
3. What state change occurs in the system?
    a. Invoice Stat: Null -> Issued -> Unpaid / Due
    b. Account Standing: Settled -> Outstanding Balance Pending.
4. Example
    a. A monthly invoice will be sent to the member's email/App; the system will assign it the Invoice number, issue date, etc. Once the member receives the invoice, they can choose between letting the system charge automatically through the default payment method, or manually pay on the app or onsite.



INVOICE

LICERIA & CO.

INVOICE NUMBER: #1234
DATE: AUGUST 12, 2030
DUE DATE: SEPTEMBER 12, 2030

123 ANYWHERE ST.,
ANY CITY
+123-456-7890

Bill To:
OLIVIA WILSON
123 Anywhere st., Any City
+123-456-7890

Payment Method
Borcelle Bank
AVERY DAVIS
+123-456-7890

DESCRIPTION	QTY	PRICE	SUBTOTAL
Premium Annual Membership (Unlimited access)	01	$1,950	$1,950
Personal Training (12 Session Package)	01	$1,350	$1,350
Nutritional Counseling & Meal Plan (Initial Assessment)	01	$350	$350
Spa Treatment (Post-Workout Ritual x2)	02	$180	$360
		TAX	$281
		GRAND TOTAL	$4,291

TERM & CONDITION
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent placerat vel sapien a ornare. Morbi faucibus nunc diam, in sodales nunc rutrum et. Aenean quis risus blandit, finibus lorem vitae, tincidunt purus.

FOR ANY QUESTIONS, PLEASE CONTACT HELLOIREALLYGREATSITE.COM OR +123-456-7890.

OLIVIA WILSON
GENERAL MANAGER

2.5 Payment & Receipt

1. Who initiates the action?
    a. Billing Engine (processes the transactions)
    b. Customer (If paying manually)
    c. Front-Desk Staff (If processing cash, POS machine, and bank transfer)
2. What data is captured or updated?
    a. Payment transaction ID, payment gateway reference code, payment method used, payment timestamp, payment status, and receipt ID.
    b. Member Account balance.
3. What state change occurs in the system?
    a. Invoice State: Unpaid / Due -> Paid / Settled
    b. Receipt State: Null -> Generated / Issued
    c. Account Standing: Payment Due -> Active Renewed
4. Example:
    a. Once a member paid his monthly invoice or any add-on service/item purchase, they will receive a receipt from the system/front-desk staff, it contains the payment transaction ID, payment gateway reference code, payment method used, etc. The member can use this receipt as evidence of the purchase and can ask for a refund if certain conditions are met.



3. Business Rules and Constraints

3.1 Membership Registration

Process:	Membership Registration
Business rules:	1. A customer can have one active membership contract only. (1:1) 2. A customer must provide only one health certificate. (1:1) 3. A member can have multiple emergency contacts. (1:M) 4. A customer must contain a unique customer ID, first name, last name, ID number, address, phone number, email address, and timestamp. 5. A registration must contain registration_ID, registration_date, registration_time, and registration_status. 6. An emergency contact is a weak entity that depends entirely on a registered member. It cannot exist without a parent member record, uses contact_name as a partial key, and is identified in combination with the member's member_id. 7. A club must contain club_ID, club_name, street_address, and multi-valued attribute amenities. 8. A health certificate must contain a unique certificate_ID, candidate_name, age, height, weight, and medical_officer_name. 9. A membership contract must contain a unique subscription_plan_ID, name, monthly_price, and commitment period.

Business Process Name: Member Registration		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A customer can hold only one active membership contract.	1:1
Backward/Passive	An active membership contract can be held by only one customer.	1:1
Conclusion	A customer can hold only one active membership contract	1:1
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A customer must provide only one health certificate.	1:1

Backward/Passive	A health certificate is provided by only one customer.	1:1
Conclusion	A customer must provide only one health certificate.	1:1
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A member can have multiple emergency contacts.	1:M
Backward/Passive	An emergency contact is associated with only one member.	1:1
Conclusion	A member can have multiple emergency contacts.	1:M
Rule No. 4		
A member must contain a unique member ID, first name, last name, ID number, address, phone number, and email.		
Rule No. 5		
A registration must contain registration_ID, registration_date, registration_time, and registration_status.		
Rule No. 6		
An emergency contact is a weak entity that depends entirely on a registered member. It cannot exist without a parent member record, uses contact_name as a partial key, and is identified in combination with the member's member_id.		
Rule No. 7		
A club must contain club_ID, club_name, street_address, and multi-valued attribute amenities.		
Rule No. 8		
A health certificate must contain a unique certificate_ID, candidate_name, age, height, weight, and medical_officer_name.		
Rule No. 9		
A membership contract must contain a unique subscription_plan_ID, name, monthly_price, and commitment period.		

3.2 Monthly Invoice

Process:	Monthly Invoice
Business rules:	1. A member can receive many monthly invoices. (1:M) 2. A monthly invoice can contain many line items. (1:M) 3. An active membership can be linked to many monthly invoices. (1: M) 4. A club branch can issue many monthly invoices. (1:M) 5. A tax rate policy can apply to many monthly invoices. (1:M) 6. Many invoices apply to many discount vouchers. (M:N) 7. A monthly invoice must contain a unique invoice_id, member_id, contract_id, club_id, issue_date, due_date, billing_cycle, tax_amount, and derived attribute total_payable_amount. 8. An invoice line item must contain line_item_ID, invoice_id, service_id, description, unit_price, quantity, and line_total. 9. A tax_rate_policy must contain a unique tax_policy_id, policy_name, and tax_type. 10. A discount voucher must contain a unique voucher_ID, code, name, and discount_rate.

Business Process Name: Monthly Invoice		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A member can receive many monthly invoices.	1:M
Backward/Passive	A monthly invoice can be issued to only one member.	1:1
Conclusion	A member can receive many monthly invoices.	1:M
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A monthly invoice can contain many line items.	1:M
Backward/Passive	A line item can belong to many monthly invoices.	1:M
Conclusion	Many monthly invoices can contain many line items.	M:N
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A monthly invoice must be linked to an active membership.	1:1
Backward/Passive	An active membership can be linked to many monthly invoices.	1:M

Conclusion	An active membership can be linked to many monthly invoices.	1:M
Rule No. 4	**Rule Statement**	**Relationship**
Forward/Active	A club branch can issue many monthly invoices.	1:M
Backward/Passive	A monthly invoice can be issued by only one club branch.	1:1
Conclusion	A club branch can issue many monthly invoices.	1:M
Rule No. 5	**Rule Statement**	**Relationship**
Forward/Active	A tax rate policy can apply to many monthly invoices.	1:M
Backward/Passive	A monthly invoice can apply only one tax rate policy.	1:1
Conclusion	A tax rate policy can apply to many monthly invoices.	1:M
Rule No. 6	**Rule Statement**	**Relationship**
Forward/Active	A monthly invoice can apply to many discount vouchers.	1:M
Backward/Passive	A discount voucher can be applied to many monthly invoices.	1:M
Conclusion	Many invoices apply to many discount vouchers.	M:N
Rule No. 7		
A monthly invoice must contain a unique invoice_id, contract_id, club_id, issue_date, due_date, billing_cycle, tax_amount, and derived attribute total_payable_amount.		
Rule No. 8		
An invoice line item must contain line_item_ID, invoice_id, service_id, description, unit_price, quantity, and line_total		
Rule No. 9		
A tax_rate_policy must contain a unique tax_policy_id, policy_name, and tax_type.		
Rule No. 10		
A discount voucher must contain a unique voucher_ID, code, name, and discount_rate.		

3.3 Payment & Receipt

Process:	Payment & Receipt
Business rules:	1. An invoice can be paid by multiple split transactions. (1:M) 2. A refund must correspond to the original transaction. (1:1) 3. A front-desk staff member can record multiple counter in-person payments. (1:M) 4. One receipt can only be referred to by one payment(1:1) 5. A Payment record should contain Payment_ID, Invoice_ID, Payment_timestamp, Amount_paid, Payment_status, Payment_method_type. 6. A cash payment record must contain cash_tendered_amount, change_returned_amount, and cashier_staff_id. 7. A card payment record must contain last_four_digits, card_network_type, gateway_transaction_ref, and authorization_code. 8. A receipt must contain a Receipt_ID, Payment_ID, issue_date, member name, business name, tax_ID, amount paid, and part of the card number.

Business Process Name: Payment & Receipt		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A transaction belongs to only one invoice.	1:1
Backward/Passive	An invoice can be paid by multiple split transactions.	1:M
Conclusion	An invoice can be paid by multiple split transactions.	1:M
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A cashier can record multiple counter in-person payments.	1:M
Backward/Passive	A counter in-person payment is recorded only by one cashier.	1:1
Conclusion	A front-desk staff member can record multiple counter in-person payments.	1:M
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	One receipt can only be referred to by one payment.	1:1
Backward/Passive	One payment can only generate one receipt.	1:1

Conclusion	Each receipt corresponds to exactly one payment, and each payment produces exactly one receipt.	1:1
Rule No. 4		
A Payment record should contain Payment_ID, Invoice_ID, Payment_timestamp, Amount_paid, Payment_status, Payment_method_type.		
Rule No. 5		
A cash payment record must contain cash_tendered_amount, change_returned_amount, and cashier_staff_id.		
Rule No. 6		
A card payment record must contain last_four_digits, card_network_type, gateway_transaction_ref, and authorization_code.		
Rule No. 7		
A receipt must contain a Receipt_ID, Payment_ID, issue_date, member name, business name, tax_ID, amount paid, and part of the card number.		
Rule No. 8		
A cashier must have a unique cashier_ID, name, gender, email, and phone_number.		

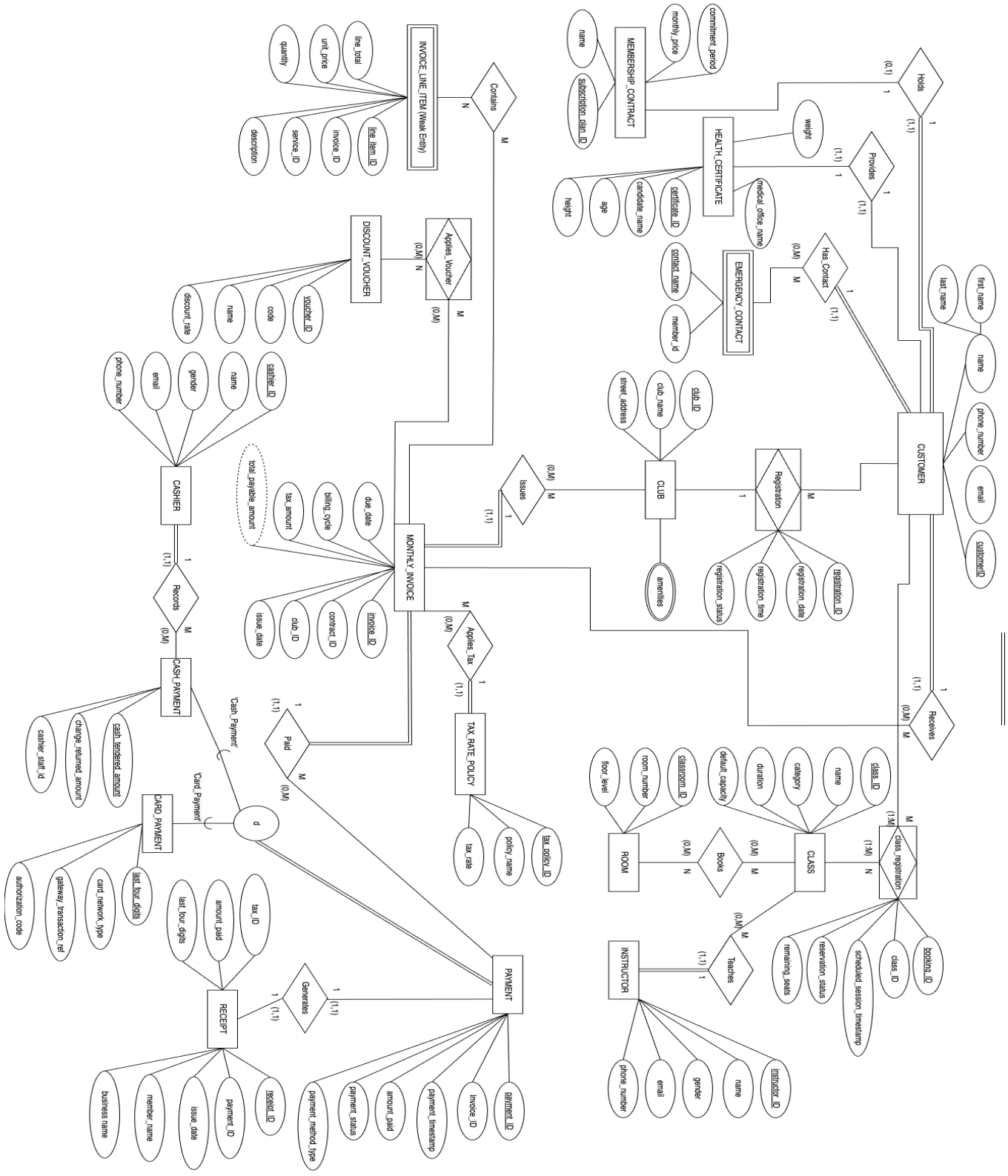
3.4 Class Registration

Process:	Class Registration
Business rules:	1. Many members can register for many classes. (M:N) 2. An instructor can teach many classes. (1:M) 3. Many classes can book many rooms. (M:N) 4. A class registration must contain Booking ID, Class ID, Instructor ID, scheduled session timestamp, member reservation status, and remaining class capacity. 5. A member can only register for one class at a time; no overlapping allowed. 6. A class must have a unique class_ID, name, category, duration, and default_capacity. 7. A classroom must have a unique classroom_ID, club_ID, room_number, and floor_level.

Business Process Name: Class Registration		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A member can register for many classes.	1:M
Backward/Passive	A class can be registered by many members.	1:M
Conclusion	Many members can register for many classes.	M:N
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	An instructor can teach many classes.	1:M
Backward/Passive	A class can only be taught by one instructor	1:1
Conclusion	An instructor can teach many classes.	1:M
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	A class can book many rooms.	1:M
Backward/Passive	A room can be booked for many classes.	1:M
Conclusion	Many classes can book many rooms	M:N

Rule No. 4
A class registration must contain Booking ID, Class ID, Instructor ID, scheduled session timestamp, member reservation status, and remaining class capacity.
Rule No. 5
An instructor must have a unique instructor_ID, name, gender, email, and phone_number.
Rule No. 6
A class must have a unique class_ID, name, category, duration, and default_capacity.
Rule No. 7
A classroom must have a unique classroom_ID, club_ID, room_number, and floor_level.

4. The Enhanced Entity Relationship Diagram



5. References

- [1] "Jetts Fitness," *Wikipedia, The Free Encyclopedia*, Wikimedia Foundation. https://en.wikipedia.org/wiki/Jetts_Fitness
- [2] Jetts Fitness Thailand, "Jetts 24 Hour Fitness Thailand," *Jetts Thailand*. https://www.jetts.co.th/en/home-en/
- [3] Jetts Fitness Thailand, "Terms and Conditions," *Jetts Thailand*. https://www.jetts.co.th/en/terms-and-conditions-en/
- [4] Jetts Fitness Thailand, "Join Online Sign-Up," *Jetts Thailand*. https://www.jetts.co.th/en/join-online-signup-en
- [5] Jetts 24 Hour Fitness UK, "Membership Terms & Conditions," *Jetts UK*. https://jetts.co.uk/membership-terms-and-conditions/
- [6] Jetts 24 Hour Fitness Thailand, "Personal Training," Jetts Thailand. https://www.jetts.co.th/en/personal-training-en/
- [7] Jetts 24 Hour Fitness Thailand, "Personal Training Pricing," Jetts App Thailand. https://jettsapp.jetts.co.th/PTpricing#:~:text=Jetts%2024%20hour%20fitness%20pricing.%20Table_content%3A%20%7C,77%2C000%20%7C%20770%20%7C%20240%20Days%20%7C
- [8] Jetts 24 Hour Fitness Thailand, "Class Timetable," Jetts Thailand. https://www.jetts.co.th/en/classt-timetable-en/z